

DATABASE MANAGEMENT SYSTEM LAB exercise

Week 1: Creating Database, Tables and Applying Constraints

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Section: 5

Experiment No: 1

Title

Creation of Database and Tables with Constraints using MySQL Workbench

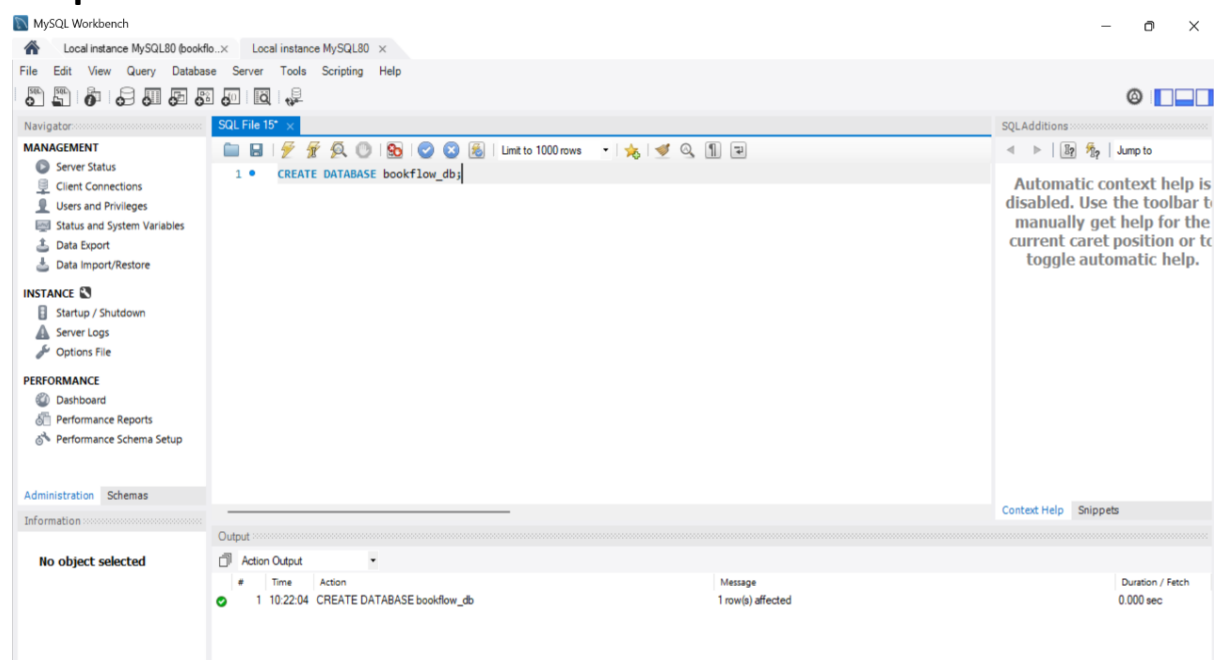
Procedure

Step 1: Create Database

Query

CREATE DATABASE IF NOT EXISTS bookflow_db;

output:

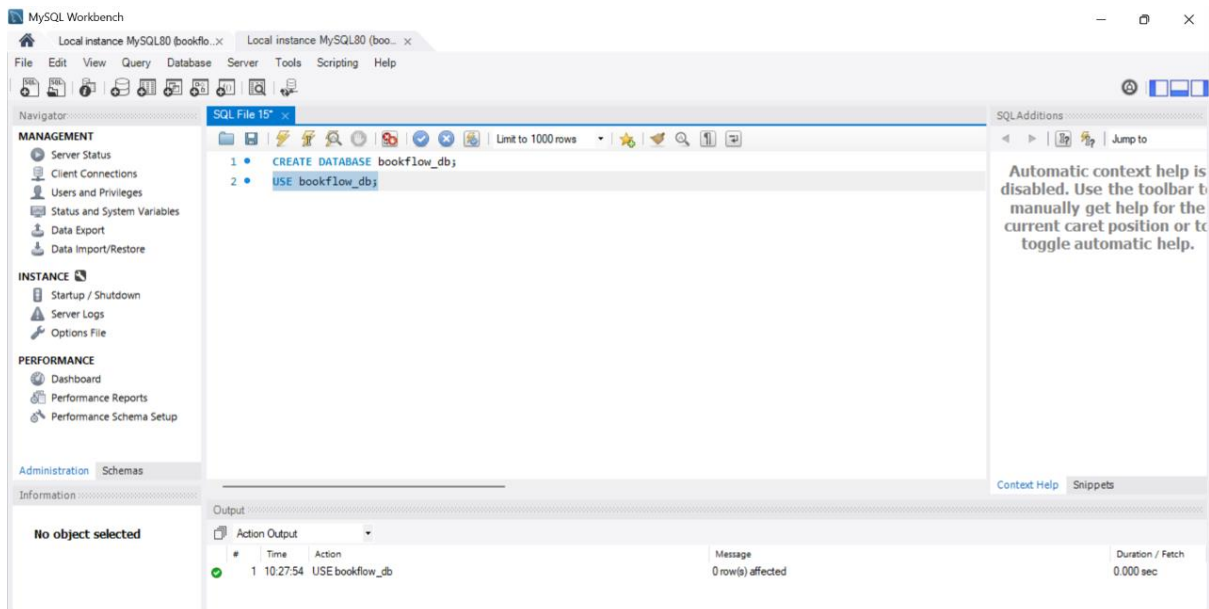


Step 2: Select the Database

Query

USE bookflow_db;

OUTPUT:

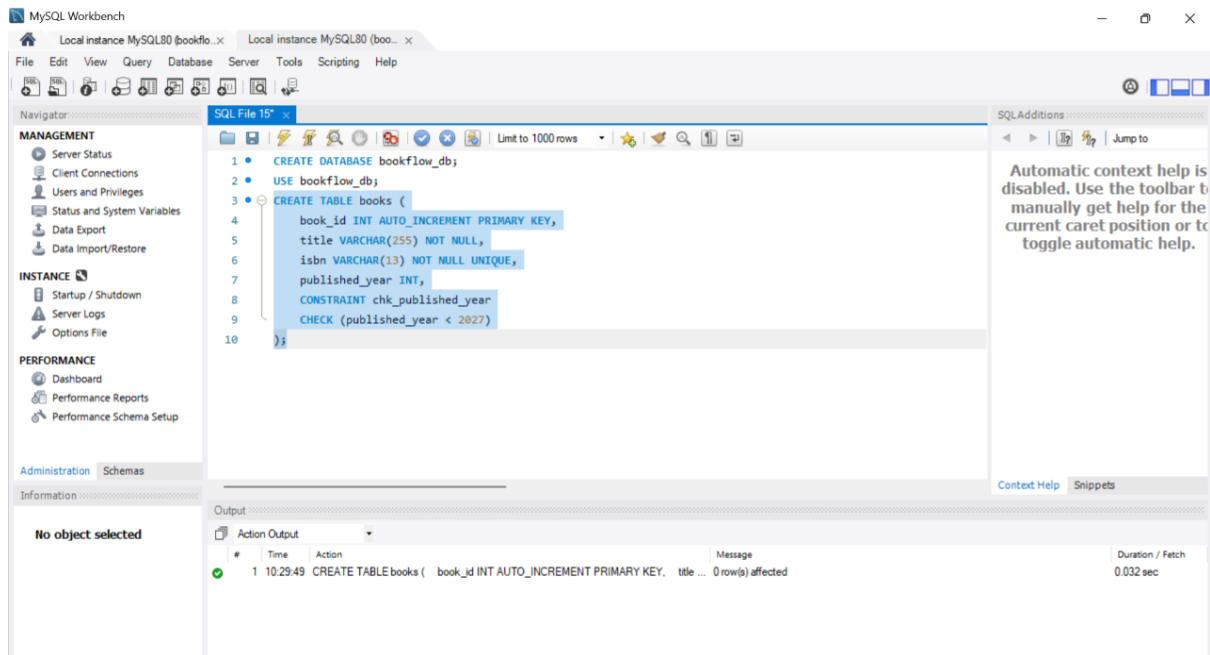


Step 3: Create Books Table

Query

```
CREATE TABLE books (  
    book_id INT AUTO_INCREMENT PRIMARY KEY,  
    title VARCHAR(255) NOT NULL,  
    isbn VARCHAR(13) NOT NULL UNIQUE,  
    published_year INT,  
    CONSTRAINT chk_published_year  
    CHECK (published_year < 2027)  
);
```

OUTPUT:

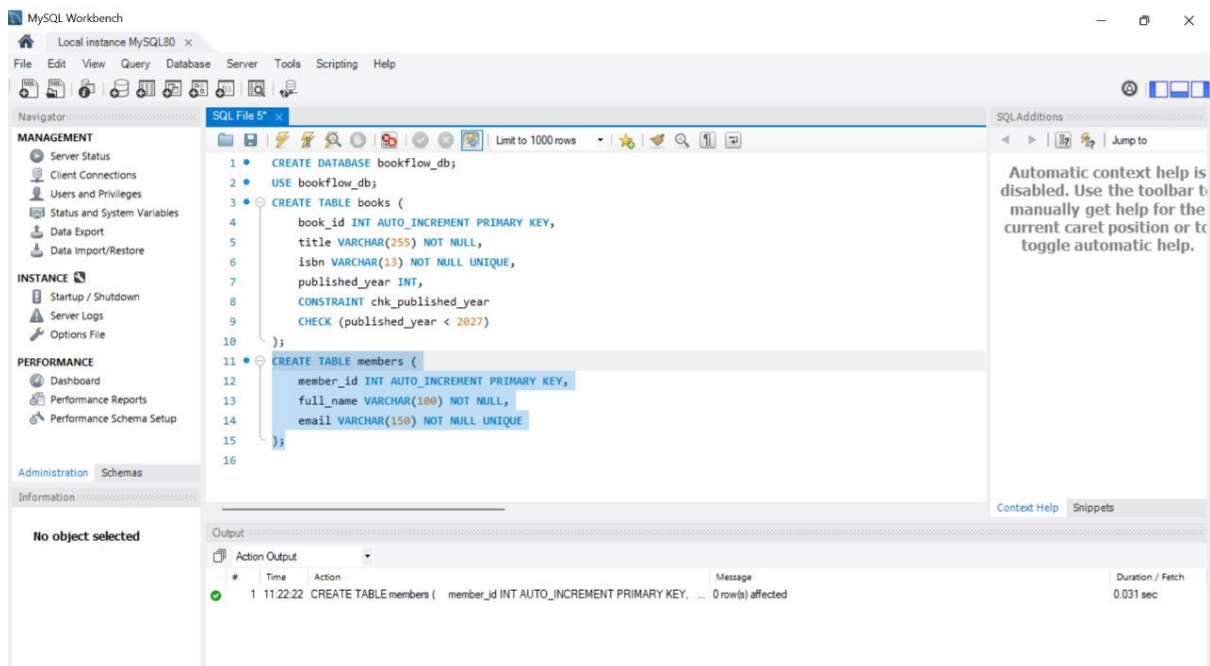


Step 4: Create Members Table

Query

```
CREATE TABLE members (  
    member_id INT AUTO_INCREMENT PRIMARY KEY,  
    full_name VARCHAR(100) NOT NULL,  
    email VARCHAR(150) NOT NULL UNIQUE  
);
```

OUTPUT:

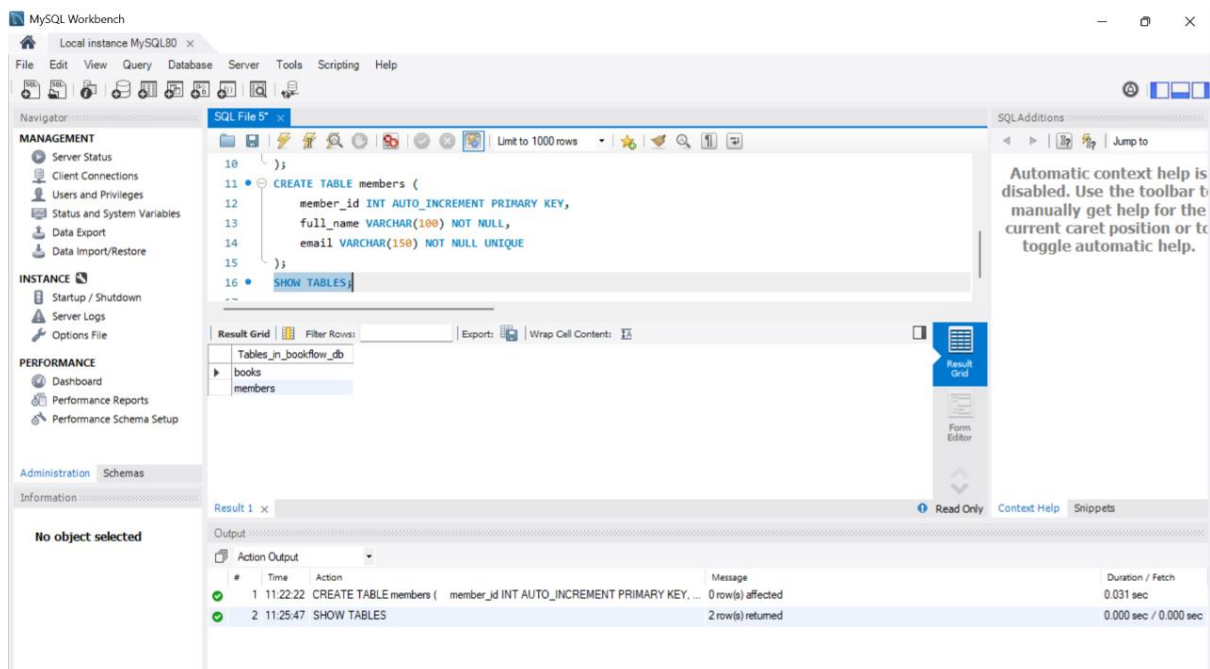


Step 5: Display Tables

Query

SHOW TABLES;

OUTPUT:



Step 6: Insert Records into Books Table

Query

INSERT INTO books (title, isbn, published_year)

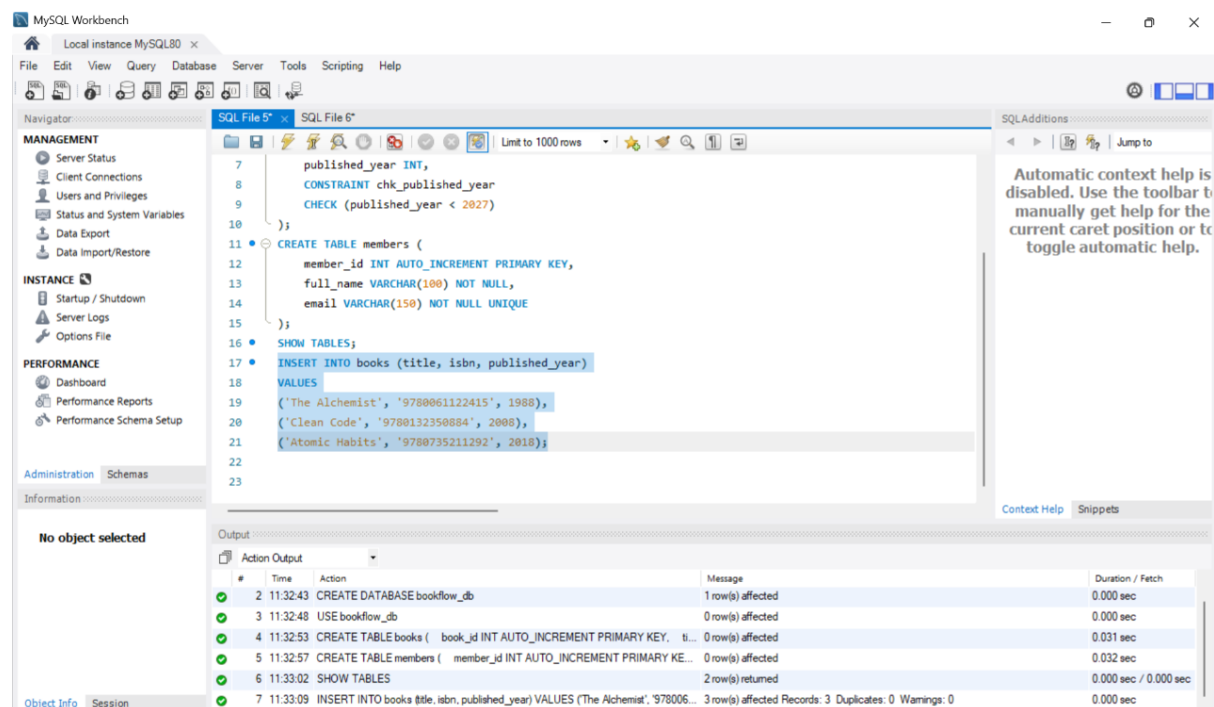
VALUES

('The Alchemist', '9780061122415', 1988),

('Clean Code', '9780132350884', 2008),

('Atomic Habits', '9780735211292', 2018);

OUTPUT:



The screenshot shows the MySQL Workbench interface. The SQL editor contains the following queries:

```
7 published_year INT,  
8 CONSTRAINT chk_published_year  
9 CHECK (published_year < 2027)  
10 );  
11 CREATE TABLE members (  
12 member_id INT AUTO_INCREMENT PRIMARY KEY,  
13 full_name VARCHAR(100) NOT NULL,  
14 email VARCHAR(150) NOT NULL UNIQUE  
15 );  
16 SHOW TABLES;  
17 INSERT INTO books (title, isbn, published_year)  
18 VALUES  
19 ('The Alchemist', '9780061122415', 1988),  
20 ('Clean Code', '9780132350884', 2008),  
21 ('Atomic Habits', '9780735211292', 2018);  
22  
23
```

The Output pane shows the results of the queries:

#	Time	Action	Message	Duration / Fetch
2	11:32:43	CREATE DATABASE bookflow_db	1 row(s) affected	0.000 sec
3	11:32:48	USE bookflow_db	0 row(s) affected	0.000 sec
4	11:32:53	CREATE TABLE books (book_id INT AUTO_INCREMENT PRIMARY KEY, ti...	0 row(s) affected	0.031 sec
5	11:32:57	CREATE TABLE members (member_id INT AUTO_INCREMENT PRIMARY KE...	0 row(s) affected	0.032 sec
6	11:33:02	SHOW TABLES	2 row(s) returned	0.000 sec / 0.000 sec
7	11:33:09	INSERT INTO books (title, isbn, published_year) VALUES ('The Alchemist', '978006...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec

Step 7: Insert Records into Members Table

Query

INSERT INTO members (full_name, email)

VALUES

('Anil Kumar', 'anil.kumar@example.com'),

('Priya Sharma', 'priya.sharma@example.com'),

('Ravi Verma', 'ravi.verma@example.com');

OUTPUT:

The screenshot shows the MySQL Workbench interface. The SQL Editor contains the following queries:

```
11 CREATE TABLE members (  
12     member_id INT AUTO_INCREMENT PRIMARY KEY,  
13     full_name VARCHAR(100) NOT NULL,  
14     email VARCHAR(150) NOT NULL UNIQUE  
15 );  
16 SHOW TABLES;  
17 INSERT INTO books (title, isbn, published_year)  
18 VALUES  
19 ('The Alchemist', '9780061122415', 1988),  
20 ('Clean Code', '9780132350884', 2008),  
21 ('Atomic Habits', '9780735211292', 2018);  
22 INSERT INTO members (full_name, email)  
23 VALUES  
24 ('Anil Kumar', 'anil.kumar@example.com'),  
25 ('Priya Sharma', 'priya.sharma@example.com'),  
26 ('Ravi Verma', 'ravi.verma@example.com');
```

The Output tab shows the results of the queries:

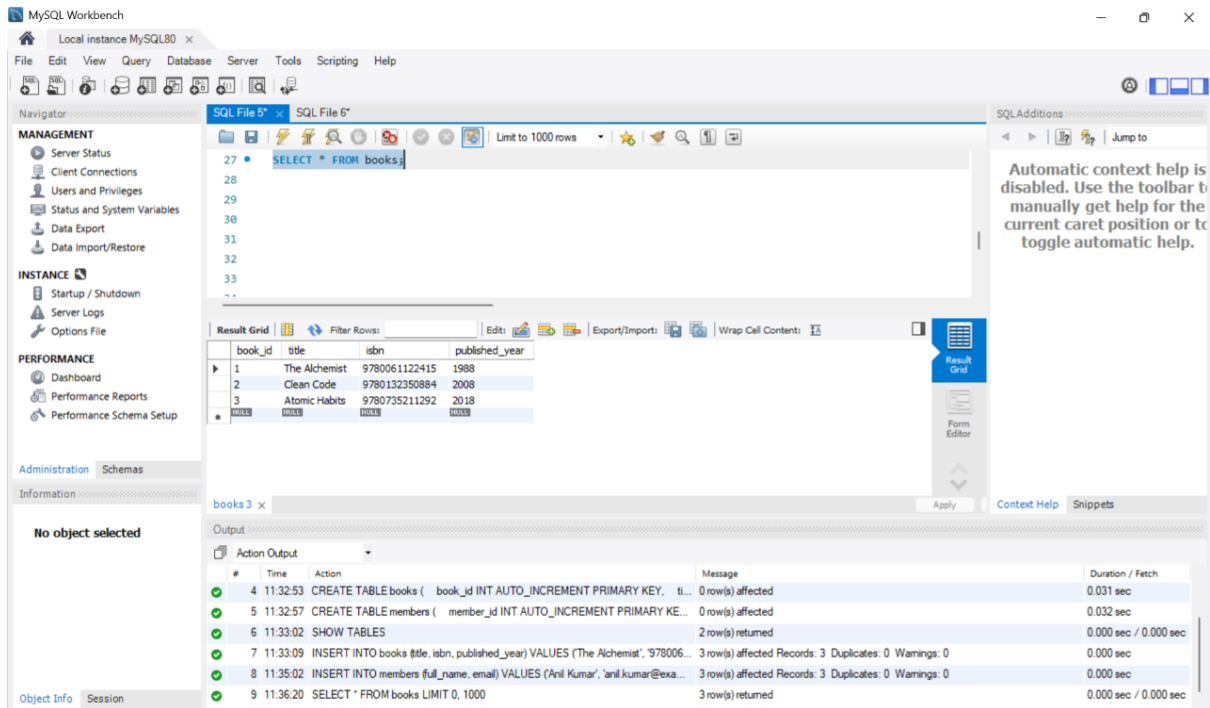
#	Time	Action	Message	Duration / Fetch
3	11:32:48	USE bookflow_db		0.000 sec
4	11:32:53	CREATE TABLE books (book_id INT AUTO_INCREMENT PRIMARY KEY, ti...	0 row(s) affected	0.031 sec
5	11:32:57	CREATE TABLE members (member_id INT AUTO_INCREMENT PRIMARY KE...	0 row(s) affected	0.032 sec
6	11:33:02	SHOW TABLES	2 row(s) returned	0.000 sec / 0.000 sec
7	11:33:09	INSERT INTO books (title, isbn, published_year) VALUES ('The Alchemist', '978006...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec
8	11:35:02	INSERT INTO members (full_name, email) VALUES ('Anil Kumar', 'anil.kumar@exa...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec

Step 8: Display Books Table

Query

SELECT * FROM books;

OUTPUT:



MySQL Workbench interface showing the output of a SQL query. The query is `SELECT * FROM books;` and the result grid displays 3 rows of data:

book_id	title	isbn	published_year
1	The Alchemist	9780061122415	1988
2	Clean Code	9780132350884	2008
3	Atomic Habits	9780735211292	2018

The Output pane shows the following actions:

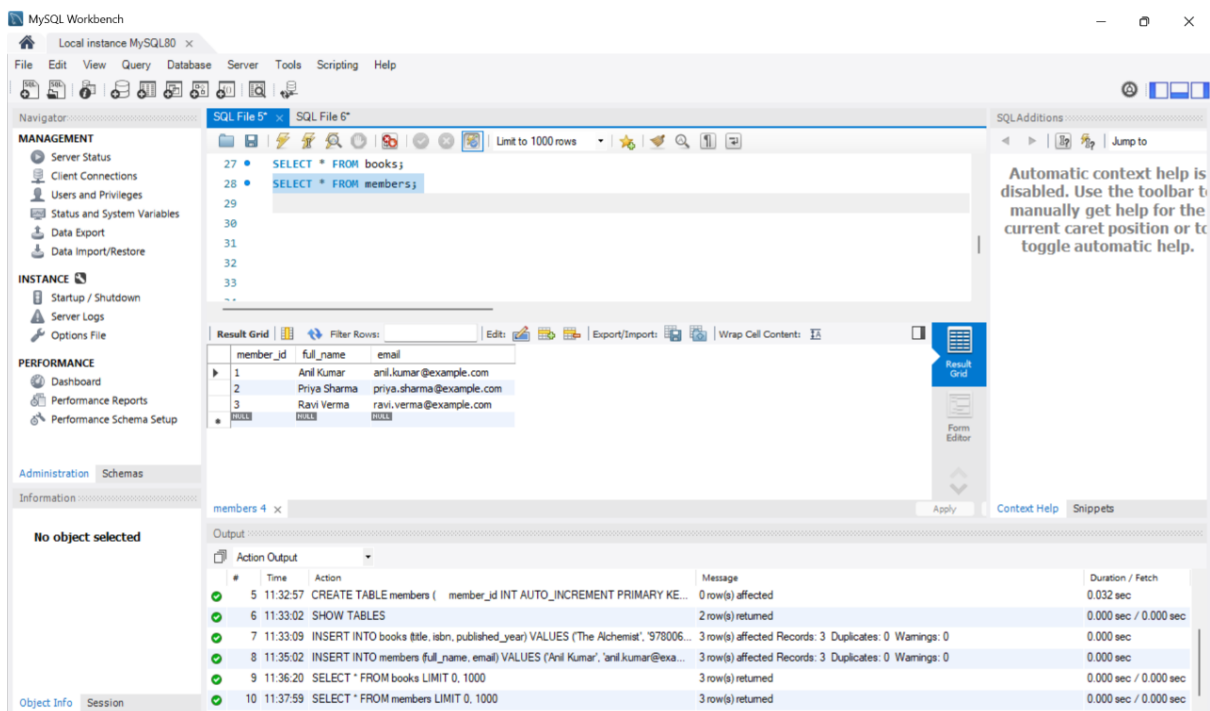
#	Time	Action	Message	Duration / Fetch
4	11:32:53	CREATE TABLE books (book_id INT AUTO_INCREMENT PRIMARY KEY, ti...	0 row(s) affected	0.031 sec
5	11:32:57	CREATE TABLE members (member_id INT AUTO_INCREMENT PRIMARY KE...	0 row(s) affected	0.032 sec
6	11:33:02	SHOW TABLES	2 row(s) returned	0.000 sec / 0.000 sec
7	11:33:09	INSERT INTO books (title, isbn, published_year) VALUES ('The Alchemist', '978006...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec
8	11:35:02	INSERT INTO members (full_name, email) VALUES ('Anil Kumar', 'anil.kumar@exa...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec
9	11:36:20	SELECT * FROM books LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec

Step 9: Display Members Table

Query

`SELECT * FROM members;`

OUTPUT:



MySQL Workbench interface showing the output of a SQL query. The query is `SELECT * FROM members;` and the result grid displays 3 rows of data:

member_id	full_name	email
1	Anil Kumar	anil.kumar@example.com
2	Priya Sharma	priya.sharma@example.com
3	Ravi Verma	ravi.verma@example.com

The Output pane shows the following actions:

#	Time	Action	Message	Duration / Fetch
5	11:32:57	CREATE TABLE members (member_id INT AUTO_INCREMENT PRIMARY KE...	0 row(s) affected	0.032 sec
6	11:33:02	SHOW TABLES	2 row(s) returned	0.000 sec / 0.000 sec
7	11:33:09	INSERT INTO books (title, isbn, published_year) VALUES ('The Alchemist', '978006...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec
8	11:35:02	INSERT INTO members (full_name, email) VALUES ('Anil Kumar', 'anil.kumar@exa...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec
9	11:36:20	SELECT * FROM books LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
10	11:37:59	SELECT * FROM members LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec